```
// C++ code
//
const int lr=5;
const int p=A0;
const int trig= 1;
const int echo= 3;
const int lg= 6;
const int ly= 9;
long distance;
int br;
float duration;
void setup()
{
  pinMode(lr, OUTPUT);
  pinMode(ly, OUTPUT);
  pinMode(lg, OUTPUT);
  pinMode(p, INPUT);
  pinMode(trig, OUTPUT);
  pinMode(echo, INPUT);
  Serial.begin(9600);
}

float mapf(float input, float ir1, float ir2, float or1, float or2)
{
  float diffi=input-ir1; //to see the difference between the input value and the start of the range
  float outrange= or2-or1; //to get the range of the output
  float inrange= ir2-ir1; //to get the range of the input
  float m= (diffi*outrange/inrange)+or1; //added or1 to get the value in the range of the output
//the final value is converted to output range so that the connected device can actually use the
value
  return m; //to send value back when the function is called
}


void loop()
{
  digitalWrite(trig, LOW);
  delayMicroseconds(2);
  digitalWrite(trig,HIGH);
  delayMicroseconds(10);
  digitalWrite(trig,LOW);

  duration= pulseIn(echo,HIGH);
  distance= 0.034*duration/2;
```

```
  br=analogRead(p);
  float i=mapf(br,0.0,1023.0,0.0,255.0);

  Serial.print("Distance: ");
  Serial.println(distance);
  Serial.print("Blink rate: ");
  Serial.println(br);

  if (distance>40&& distance<331)
  {
    digitalWrite(lg,HIGH);
    digitalWrite(ly,LOW);
    digitalWrite(lr,LOW);
    analogWrite(lg,i);
    Serial.println("Green LED is on");
  }
  else if(distance>15&&distance<=40)
  {
    digitalWrite(ly,HIGH);
    digitalWrite(lg,LOW);
    digitalWrite(lr,LOW);
    analogWrite(ly,i);
    Serial.println("Yellow LED is on");
  }
  else if(distance<15)
  {
    digitalWrite(lr,HIGH);
    digitalWrite(ly,LOW);
    digitalWrite(lg,LOW);
    analogWrite(lr,i);
    Serial.println("Red LED is on");
  }
  else
  {
   Serial.println("Outside sensor range");
   Serial.println("No LED is on");
  }
}
```